National Testing Agency

Question Paper Name :	B Tech 6th Apr 2026 Shift 2
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B. Tech

Group Number :	1
Group Id :	69112117
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	69112197
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112197
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 6911211201 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f: \mathbf{R} \to \mathbf{R}$ be defined as $f(x) = \dfrac{2x^2 - 3x + 2}{3x^2 + x + 3}$. Then f is :

Options :

6911214081. both one-one and onto

6911214082. one-one but not onto

6911214083. onto but not one-one

6911214084. neither one-one nor onto

Question Number : 2 Question Id : 6911211202 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the quadratic equation $(n^2-2n+2)x^2-3x+(n^2-2n+2)^2=0$, $n \in \mathbf{R}$. Let α be the minimum value of the product of its roots and β be the maximum value of the sum of its roots. Then the sum of the first six terms of the G.P., whose first term is α and the common ratio is $\frac{\alpha}{\beta}$, is :

Options :

6911214085. $\frac{61}{37}$

6911214086. $\frac{121}{81}$

6911214087. $\frac{364}{243}$

6911214088. $\frac{1093}{729}$

Question Number : 3 Question Id : 6911211203 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $S=\left\{z \in \mathbb{C} : z^2+\sqrt{6}\,iz-3=0\right\}$. Then $\sum_{z \in S} z^8$ is equal to :

Options :

6911214089. 162

6911214090. 184

6911214091. 262

6911214092. 324

Question Number : 4 Question Id : 6911211204 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The sum of all possible values of $\theta \in [0, 2\pi]$, for which the system of equations :

$$x\cos 3\theta - 8y - 12z = 0$$
$$x\cos 2\theta + 3y + 3z = 0$$
$$x + y + 3z = 0$$

has a non-trivial solution, is equal to :

Options :

6911214093. π

6911214094. 2π

6911214095. 3π

6911214096. 4π

Question Number : 5 Question Id : 6911211205 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A=\begin{bmatrix}1 & 0 & 0\\ 3 & 1 & 0\\ 9 & 3 & 1\end{bmatrix}$ and $B=[b_{ij}]$, $1\le i, j\le 3$. If $B=A^{99}-I$, then the value of $\frac{b_{31}-b_{21}}{b_{32}}$ is :

Options :

6911214097. 99

6911214098. 199

6911214099. 149

6911214100. 159

Question Number : 6 Question Id : 6911211206 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The sum $1+\frac{1}{2}\left(1^2+2^2\right)+\frac{1}{3}\left(1^2+2^2+3^2\right)+\ldots.$ upto 10 terms is equal to :

Options :

6911214101. 130

6911214102. 155

6911214103. $\frac{315}{2}$

6911214104. $\frac{325}{2}$

Question Number : 7 Question Id : 6911211207 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A building has ground floor and 10 more floors. Nine persons enter in a lift at the ground floor. The lift goes up to the 10th floor. The number of ways, in which any 4 persons exit at a floor and the remaining 5 persons exit at a different floor, if the lift does not stop at the first and the second floors, is equal to :

Options :

6911214105. 2184

6911214106. 3064

6911214107. 7056

6911214108. 11340

Question Number : 8 Question Id : 6911211208 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the mean and the variance of seven observations $2, 4, \alpha, 8, \beta, 12, 14, \alpha < \beta$, be 8 and 16 respectively. Then the quadratic equation whose roots are $3\alpha+2$ and $2\beta+1$ is :

Options :

6911214109. $x^2-35x+306=0$

6911214110. $x^2-41x+420=0$

6911214111. $x^2-45x+506=0$

6911214112. $x^2-37x+342=0$

Question Number : 9 Question Id : 6911211209 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A bag contains 6 blue and 6 green balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that each drawn pair consists of one blue and one green ball is :

Options :

6911214113. $\frac{63}{925}$

6911214114. $\frac{17}{231}$

6911214115. $\frac{16}{231}$

6911214116. $\frac{64}{925}$

Question Number : 10 Question Id : 6911211210 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let C be a circle having centre in the first quadrant and touching the x-axis at a distance of 3 units from the origin. If the circle C has an intercept of length $6\sqrt{3}$ on y-axis, then the length of the chord of the circle C on the line $x-y=3$ is :

Options :

6911214117. 8

6911214118. 6

6911214119. $6\sqrt{2}$

6911214120. $8\sqrt{2}$

Question Number : 11 Question Id : 6911211211 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x = \pm \frac{4\sqrt{6}}{3}$.

Let $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be a hyperbola whose eccentricity is equal to the length of semi-major axis of E, and whose length of latus rectum is equal to the length of minor axis of E. Then the distance between the foci of H is :

Options :

6911214121. $\frac{4\sqrt{2}}{\sqrt{7}}$

6911214122. $\frac{4\sqrt{2}}{7}$

6911214123. $\frac{4}{\sqrt{7}}$

6911214124. $\frac{8}{7}$

Question Number : 12 Question Id : 6911211212 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $x = 9$ be a directrix of an ellipse E, whose centre is at the origin and eccentricity is $\frac{1}{3}$. Let P $(\alpha, 0)$, $\alpha > 0$, be a focus of E and AB be a chord passing through P. Then the locus of the mid point of AB is :

Options :

6911214125. $9y^2 = 8x\,(1-x)$

6911214126. $3y^2 = 4x\,(1-x)$

6911214127. $9y^2 = 8x\,(x-1)$

6911214128. $3y^2 = 4x\,(x-1)$

Question Number : 13 Question Id : 6911211213 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\sin\left(\tan^{-1}(x\sqrt{2})\right) = \cot\left(\sin^{-1}\sqrt{1-x^2}\right)$, $x \in (0, 1)$, then the value of x is :

Options :

6911214129. $\frac{1}{2}$

6911214130. $\frac{1}{3}$

6911214131. $\frac{2}{3}$

6911214132. $\frac{5}{8}$

Question Number : 14 Question Id : 6911211214 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The shortest distance between the lines $\frac{x-4}{1} = \frac{y-3}{2} = \frac{z-2}{-3}$ and $\frac{x+2}{2} = \frac{y-6}{4} = \frac{z-5}{-5}$ is :

Options :

6911214133. $\frac{5\sqrt{6}}{6}$

6911214134. $2\sqrt{5}$

6911214135. $3\sqrt{5}$

6911214136. $4\sqrt{5}$

Question Number : 15 Question Id : 6911211215 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\vec{a} = 2\hat{i} + 3\hat{j} + 3\hat{k}$ and $\vec{b} = 6\hat{i} + 3\hat{j} + 3\hat{k}$. Then the square of the area of the triangle with adjacent sides determined by the vectors $\left(2\vec{a} + 3\vec{b}\right)$ and $\left(\vec{a} - \vec{b}\right)$ is :

Options :

6911214137. 450

6911214138. 900

6911214139. 1800

6911214140. 2400

Question Number : 16 Question Id : 6911211216 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\lim\limits_{x \to 2} \frac{(\tan(x-2))\left(rx^2+(p-2)x-2p\right)}{(x-2)^2} = 5$ for some r, p ϵ **R**. If the set of all possible values of q, such that the roots of the equation $rx^2 - px + q = 0$ lie in (0, 2), be the interval $(\alpha, \beta]$, then $4(\alpha+\beta)$ equals :

Options :

6911214141. 11

6911214142. 13

6911214143. 17

6911214144. 21

Question Number : 17 Question Id : 6911211217 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 1 & \alpha \\ 0 & 1 & -1 \end{bmatrix}$ be a singular matrix. Let $f(x) = \int_0^x \left(t^2 + 2t + 3\right) dt$, $x \epsilon [1, \alpha]$. If M and m are respectively the maximum and the minimum values of f in $[1, \alpha]$, then $3(M-m)$ is equal to :

Options :

6911214145. 64

6911214146. 68

6911214147. 72

6911214148. 76

Question Number : 18 Question Id : 6911211218 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f: \mathbf{R} \to \mathbf{R}$ be such that $f(xy) = f(x)f(y)$, for all $x, y \in \mathbf{R}$ and $f(0) \neq 0$. Let $g : [1, \infty) \to \mathbf{R}$ be a differentiable function such that

$$x^2 g(x) = \int_1^x \left(t^2 f(t) - t g(t)\right) dt.$$

Then g(2) is equal to :

Options :

6911214149. $\frac{13}{8}$

6911214150. $\frac{11}{16}$

6911214151. $\frac{15}{32}$

6911214152. $\frac{17}{64}$

Question Number : 19 Question Id : 6911211219 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The area of the region $\{(x, y) : x^2 - 8x \leq y \leq -x\}$ is :

Options :

6911214153. $\frac{343}{6}$

6911214154. $\frac{637}{6}$

6911214155. $\frac{437}{6}$

6911214156. $\frac{523}{6}$

Question Number : 20 Question Id : 6911211220 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The value of the integral $\int_{-1}^{1} \left(\frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} \right) dx$ is equal to :

Options :

6911214157. $3 \log_e 2$

6911214158. $2\log_e 2$

6911214159. $5\log_e 3$

6911214160. $3\log_e 3$

Mathematics Section B

Section Id :	69112198
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112198
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 6911211221 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $R = \{(x, y) \in \mathbf{N} \times \mathbf{N} : \log_e (x + y) \leq 2\}$. Then the minimum number of elements, required to be added in R to make it a transitive relation, is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 22 Question Id : 6911211222 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $(1-x^3)^{10} = \sum_{r=0}^{10} a_r x^r (1-x)^{30-2r}$, then $\frac{9a_9}{a_{10}}$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 23 Question Id : 6911211223 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let the line $x - y = 4$ intersect the circle $C : (x-4)^2 + (y+3)^2 = 9$ at the points Q and R. If $P(\alpha, \beta)$ is a point on C such that $PQ = PR$, then $(6\alpha + 8\beta)^2$ is equal to _____.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 6911211224 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let the image of the point $P(0,-5,0)$ in the line $\frac{x-1}{2}=\frac{y}{1}=\frac{z+1}{-2}$ be the point R and the image of the point $Q\left(0,\frac{-1}{2},0\right)$ in the line $\frac{x-1}{-1}=\frac{y+9}{4}=\frac{z+1}{1}$ be the point S. Then the square of the area of the parallelogram PQRS is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 6911211225 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $f(x)=\begin{cases}x^3+8; & x<0,\\ x^2-4; & x\geq 0,\end{cases}$ and $g(x)=\begin{cases}(x-8)^{1/3}; & x<0,\\ (x+4)^{1/2}; & x\geq 0.\end{cases}$

Then the number of points, where the function *gof* is discontinuous, is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id :	69112199
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112199
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 6911211226 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical